

CITC-2190 Capstone Project – Olivia Smith – Lightsaber Builder – Spring 2025

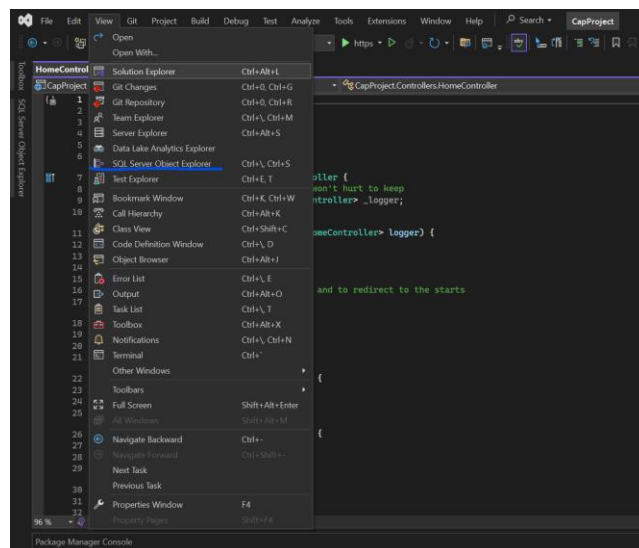
Please refer to this document before running to ensure it is loaded in the correct manner so issues can be avoided

I. Running the application

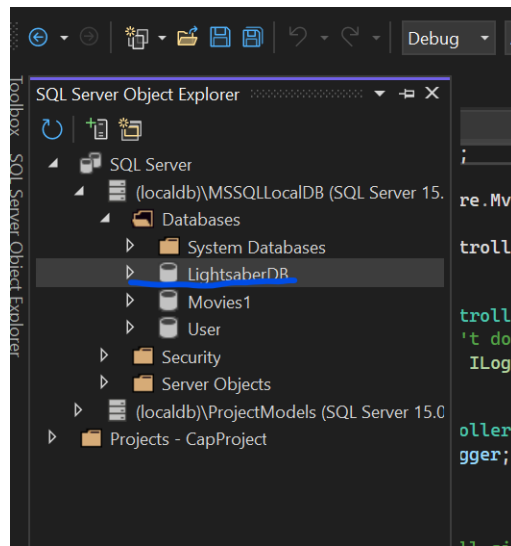
1. Please do NOT build the solution. Do not run “clean solution” or “build” in any form. This will cause Visual Studio to “lock out” the entire folder due to permission problems.
2. Navigate to the following sub folder of the solution (below) and run the “CapProject.sln” in Visual Studio 2022

Name	Date modified	Type	Size
.git	4/11/2025 9:01 PM	File folder	
.vs	4/1/2025 11:38 AM	File folder	
CapProject	4/28/2025 9:59 PM	File folder	
node_modules	4/11/2025 9:23 PM	File folder	
.gitattributes	4/11/2025 9:01 PM	Git Attributes Source ...	3 KB
.gitignore	4/11/2025 9:01 PM	Git Ignore Source File	7 KB
CapProject.sln	2/27/2025 7:40 PM	Visual Studio Solution	1 KB
package.json	4/11/2025 9:40 PM	JSON File	1 KB
package-lock.json	4/11/2025 9:41 PM	JSON File	1 KB
README.md	4/11/2025 9:01 PM	Markdown Source File	1 KB
SQLQuery1.sql	4/8/2025 3:17 PM	Microsoft SQL Server...	1 KB

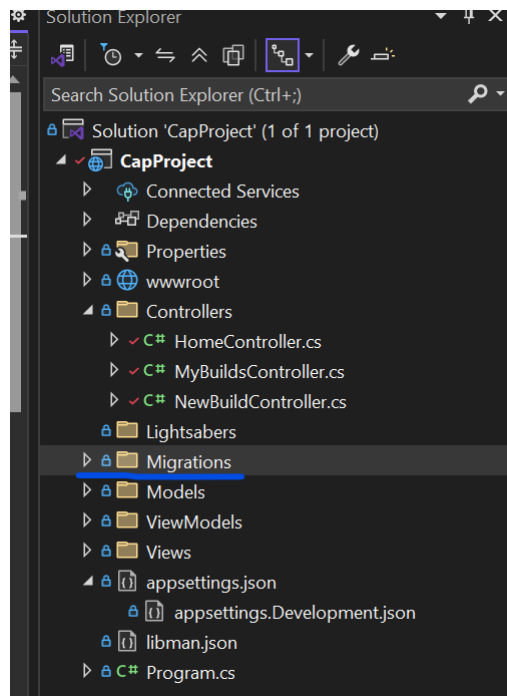
3. Once the project is open, do not run it yet. You must first rebuild the database and migration
 - a. Navigate to the View tab and select “SQL Server Object Explorer”



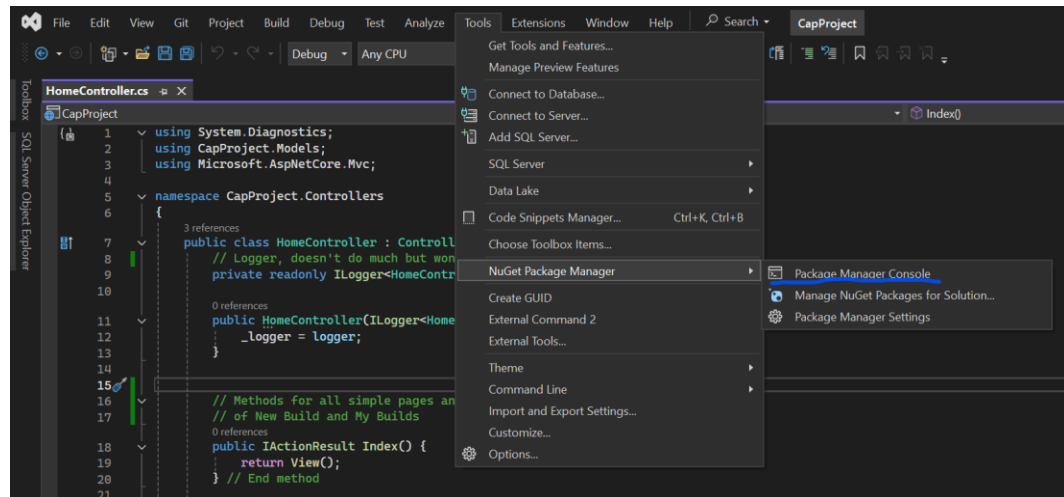
- b. Navigate through the dropdowns as shown below. If there is a database named “LightsaberDB,” right-click on it and delete the database



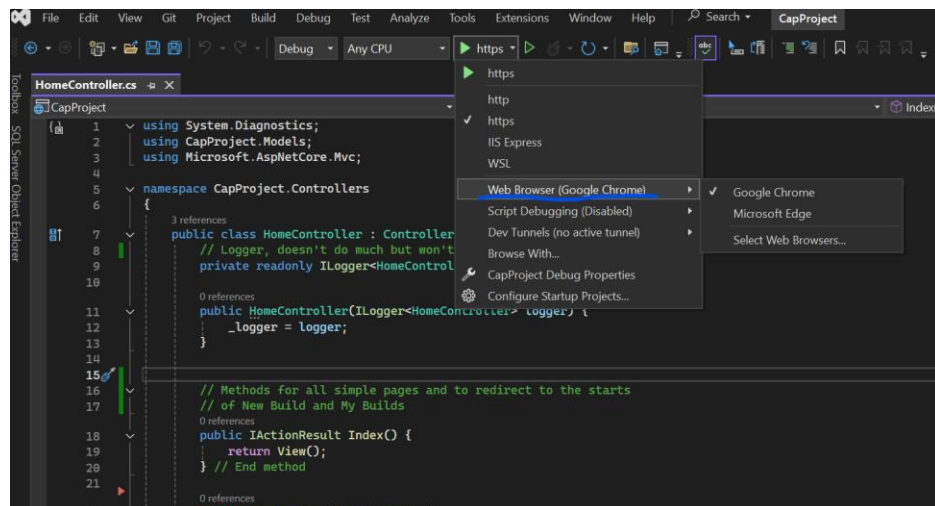
- c. If the database wasn't present, move to step 4. If it was, right-click on the Migrations folder in Solution Explorer and delete it as well as shown



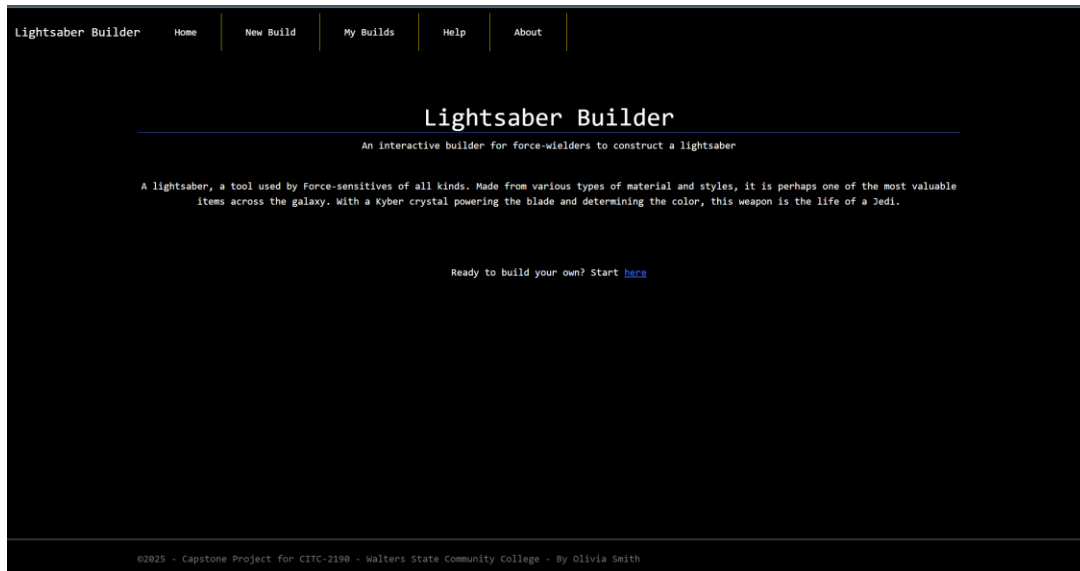
- 4. Once this is done, the database and migrations can be rebuilt
 - a. Navigate to the Tools > NuGet Package Manager > Package Manager Console



- b. In the NuGet Console, run the following commands in this order:
 - i. PM> Add Migration-Initial
 - ii. PM> Update-Database
 - c. After this, the database and migrations have been successfully rebuilt
5. Run the program by selecting the “https” button at the top
 - a. If an error occurs at first with connect issues, run in another browser or accept the risk
 - b. You may select a browser of your choice by selecting the dropdown next to the “https” button, and then web browser



6. The application should now run as needed and look as shown below



II. About the Project

This project is organized using the ASP.NET Core Model-View-Controller (MVC) Framework with C# Razor code. It also includes the following:

- HTML/CSS
- JavaScript
 - Three.JS
- Git (only for backup purposes and testing)
- SQL

The format of MVC are as follows:

- Web pages (HTML/CSS/JS/Razor) are included in the Views folder
- Contextual classes for database functions and other are included in the Models folder (C#
 - Additionally for certain data transfers in the webpages, a ViewModel folder holds a file for better functionality with this
- Files that controller redirections and some error handlings are included in the Controllers folder (C#)

Additional folders and files are as follows:

- The Migrations folder holds database related code
- The wwwroot folder contains the following:
 - Basic VS function folders (css, favicon.ico)
 - The js folder for all JavaScript files (including Three.JS)
 - The lib folder for popper.js and Bootstrap files

III. Functions and Methods used

With the languages and tools mentioned previously, this project does CRUD operations (Create, Update, Read, and Delete) using a local database. The user will go through the following prompts in creating an object:

- Enter a name
- Select an Emitter component
- Select a Switch component
- Select a Hilt component
- Select a Pommel component
- Select a color of the blade

Together, these prompts create a new Lightsaber object in the database that the user can retrieve to edit or delete as they wish.

Images of each page will be shown in section V.

In the Controllers: The user's inputs for each prompt are stored in sessions created by the application for access between pages. In both the creating and editing version of this, data is taken from the database and stored in the ViewModel for easier access/fallback options to ensure data is correct. Helper methods for populating dropdown menus are used as well.

In the Models: Variables are defined in Lightsaber.cs and Component.cs for fields in the database. To allow combinations of these two entities for better database functionality, there is the LightsaberComponent.cs file. For database implementation and seed data, there is the LightsaberContext.cs file.

In the Views: There are three files, Home, NewBuilds, and MyBuilds. Each of these folders corresponds to their own controller in the Controllers folder. Home holds simple pages such as home, help, and about. NewBuild holds all pages needed to create a Lightsaber object. The MyBuilds folder holds edited versions of the NewBuild pages along with a page to view all builds and options with them.

IV. Developer's Note

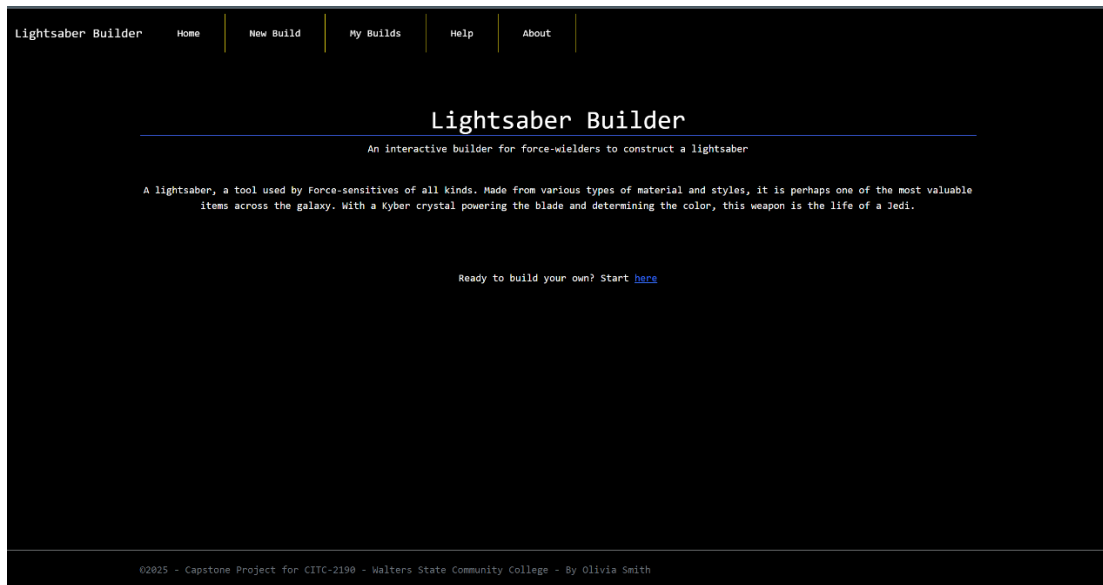
I will be the first to admit that this project is far from perfect in many ways. This application has bugs and errors, some of which I may not even know about. However, using what I have learned over the past two years has well equipped me for the progress that I've made this semester.

I plan to update and improve this application over time, with more features in mind. Regardless of what the outcome will be, this semester's work will be the foundation for the improvements that will make to this project and many others in the future.

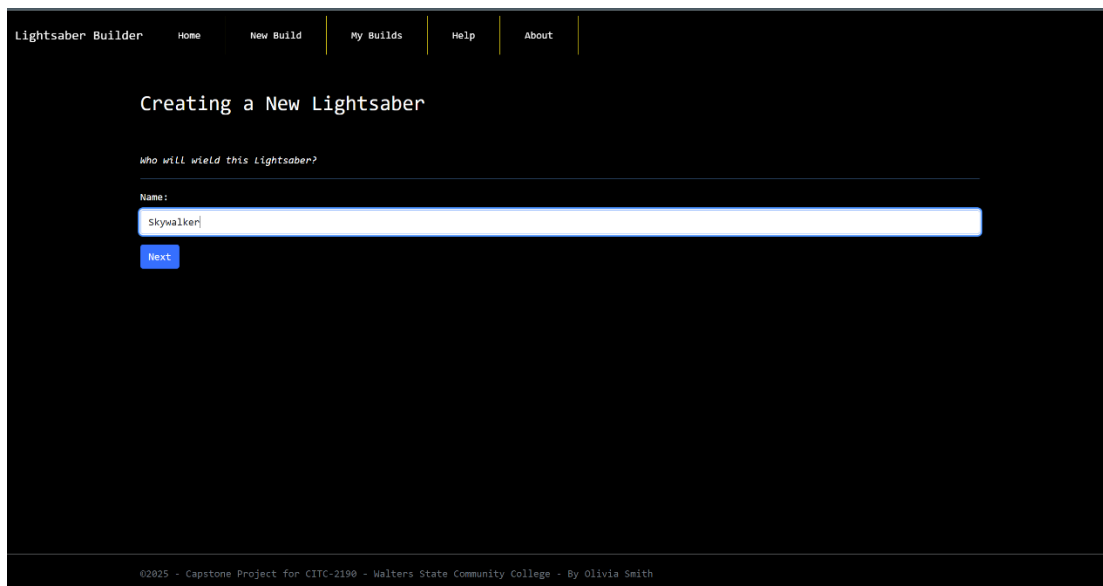
I have been fond of both coding and the *Star Wars* Universe since high school and will continue to be for many years. This project is the outcome of that fondness. I hope you enjoy it as much as I've enjoyed making it.

V. Screenshots

Home page:



New Build > Name:



New Build > Emitter:

Lightsaber Builder

Home

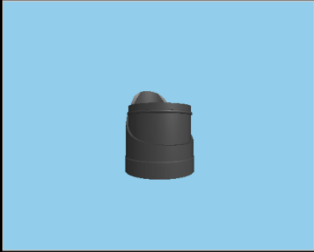
New Build

My Builds

Help

About

Choose Your Emitter



The gateway for the blade to emit from the Kyber crystal.

Emitter

Elemental Nature I

Next

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My Builds:

Lightsaber Builder

Home

New Build

My Builds

Help

About

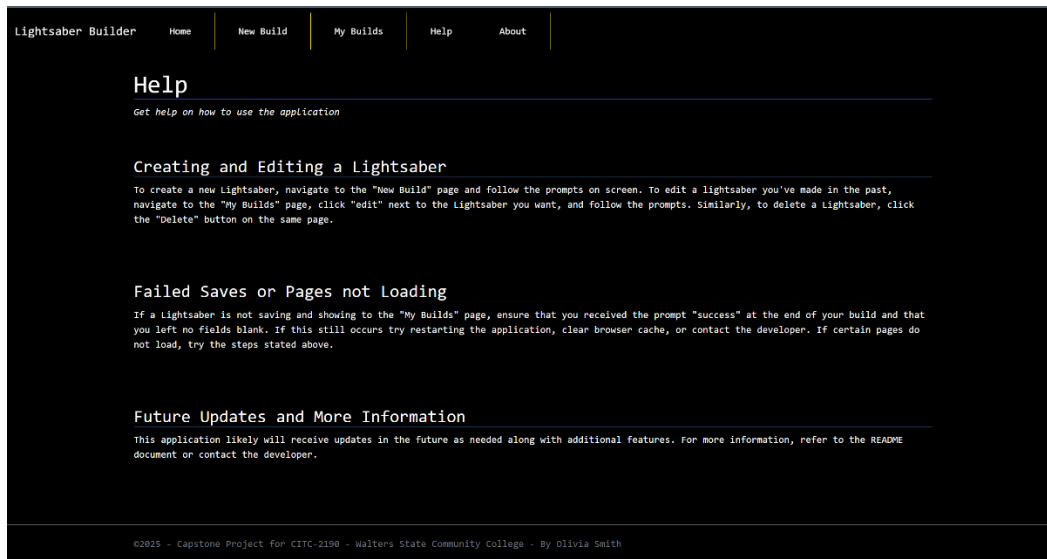
My Builds

Manage your previous builds, or remove them.

Name
Example Edit Delete

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Help:



About:

